

PROCESSES TO CONCEPTS IN TWAU CURRICULUM: DEVELOPING PROCESS SKILLS IN CLASSROOM

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4.1 INTRODUCTION

In the previous unit, we have discussed and understood that **The World Around Us** (TWAU) curriculum is interdisciplinary and entails children exploring phenomena and processes around them in a holistic manner. Children are actively engaged in constructing knowledge as discussed in Chapter 2. As the curriculum is exploratory in nature, children learn better by doing than by reading facts and concepts. The curricular goals, competencies and the learning outcomes of the 'The World Around Us', are achieved by using certain processes. These processes used for exploring and developing a conceptual understanding are called 'process skills'. School children use process skills to investigate in a manner similar to the way scientists conduct their inquiry.

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Children use process skills to construct knowledge by asking questions, making observations, collecting data, taking measurements, inferencing data, predicting the outcomes of manipulating one variable, designing investigations and communicating their findings to others. Hence, process skills form the core of inquiry based, hands-on learning in primary classrooms. To apply various processes, like any other skill, children have to first master them. The more the process skills are developed, the more they can develop a better understanding of the physical, biological and socio-cultural aspects of the environment. In this chapter, we discuss the importance of various process skills in primary classrooms and how a teacher can develop and promote these process skills in young children.

**4.2 LEARNING OUTCOMES**

After completion of this chapter, the learner

- appreciates the scope of process skills in 'The World Around Us' Curriculum;
- identifies various process skills in the given activities;
- designs activities to promote various process skills in the classroom; and
- develops strategies for developing process skills in children.

4.3 WHAT ARE PROCESS SKILLS?

Process skills refer to a range of abilities related to making observations, identifying questions, making predictions, designing investigations, interpreting data, and communicating the results. These skills are a hallmark of scientific method. Scientific method here refers to systematic exploration or investigation which is not limited to the Science subject only. Process skills are the route to conceptual understanding i.e. children can make better understanding of concepts through process skills.

Figure 4.1 illustrates some basic process skills which are useful at the foundational and preparatory stages of school in the context of teaching -learning of the 'The World Around Us'

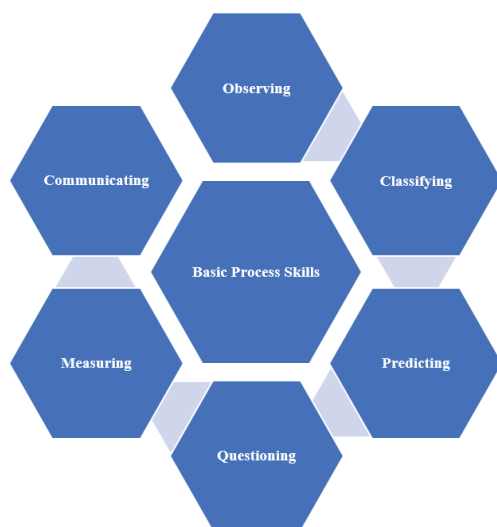


Fig. 4.1: Some Basic Process Skills Useful at Foundational and Preparatory Stages

The following sections present the meaning and scope of these process skills and how they can be developed in primary classrooms for effective teaching - learning of 'The World Around Us'.

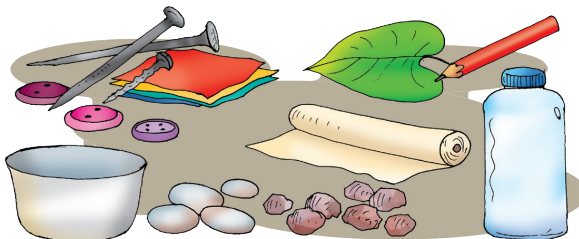
4.3.1 From Activity to Process skills

Let us understand the scope of these process skills in a primary classroom through the following activity and identify the process skills involved.

Anita teaches at Class 4. She conducts the following activity from the NCERT text-book 'Our Wondrous World' (Activity - Page No. 110 of Class 4).

Activity 3

Collect objects, such as a leaf, an iron nail, an empty bowl (*katori*), a plastic bottle with a lid, pieces of stone, aluminium foil and other things of your choice. Fill a bucket with water. Before dropping the things listed in the table, guess whether things will float or sink. You may try other things too. Then, observe what happens when it is actually dropped in water. You can write 'F' for objects that float and 'S' for objects that sink.



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Let's reflect-

- 1) Which process skills are being used by children in this activity?
- 2) In this activity, the teacher is encouraging the children to guess before performing the task. Why do you think she or he is doing so and which process skill do you think the teacher is encouraging here?

Let us discuss this activity in light of the process skills. Here the teacher is asking the children to **observe** what happens when a particular object is put in water and to note down their observation. Hence, the children are using the skill of **observation** and when they are listing their observations in a table, they are **communicating** their observations. When the teacher is asking the children to guess, she is encouraging them to use the process skill of **predicting**.

Anita further asks children to categorise the given objects into two groups - floaters and sinkers. By asking to do so, she is encouraging the process skill of **classifying**.

In a discussion which followed up the activity, some children shared that heavy objects like iron nails sink and lighter ones like leaves float. Anita discusses this further as follows:

Anita (Teacher) - Did all the heavy objects sink and all the lighter ones float?

Israr (Student) - No mam, I observed that _____ did not float

Anita (Teacher) - Name the heavy objects which floated

From the discussion, the learners **inferred** that floating or sinking cannot be decided based on heaviness only. It may depend on other factors.

The teacher further extended this activity and engaged the learners in investigating whether shape plays a role in floating and sinking. Here, the teacher is encouraging **hypothesising**.

The teacher and learners perform the activity to check their hypothesis. This activity is described on Page 112 of the textbook of Class 4.

**Activity 4**

Let us find out whether shape plays a role in floating and sinking.

Aluminium foil when—	Do they float (F) or sink (S) in water?			
	Before dropping		After dropping	
	What is your guess?	Why do you think so?	What is your observation?	What could be the reason?
spread out				
pressed tightly into a ball				
in a cup-like shape				



By changing the shape, we can make an object float or sink. When aluminium foil was spread out or made in the shape of a cup, it floated. However, when it was made into a ball, it sank.

Note to the Teacher

You need to help students in pressing the aluminium foil ball tightly so that it traps as little air as possible. If the ball is not pressed tightly, it might even float. If aluminium foil is not available, use any other alternative.

After performing this activity, the children concluded that by changing the shape we can make an object float or sink (Inferencing).

Thus, we have seen how an activity can be used to develop various process skills in children. The next section discusses in detail about these process skills.

**CHECK YOUR PROGRESS 4.1**

- 1) What are the various process skills which can be used in primary classrooms?
- 2) What is the importance of process skills in the teaching-learning of 'The World Around Us'?

4.4 DEVELOPING PROCESS SKILLS IN PRIMARY SCHOOL CHILDREN

In this section, we discuss the meaning and scope of various process skills and how they can be developed in young children through various activities.



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4.4.1 Observing

Observation or observing as a process skill involves gathering information using any of our senses. Human beings are natural observers and this basic process skill has helped us to evolve. Observation plays a very important role in attainment of developmental milestones in young children. It also assists in learning and making sense of the world around us.

Let's understand what observing involves as a process skill through the following activity given on Page 50 of the textbook of Class 4.



Activity 5

- During the visit to the nature trail, students collected leaves from different plants. Help them identify the plants based on the pictures of the leaves given below.



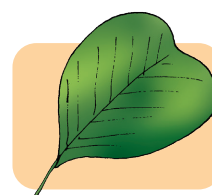
(a)



(b)



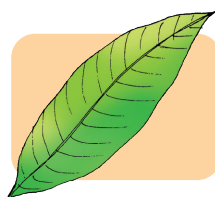
(c)



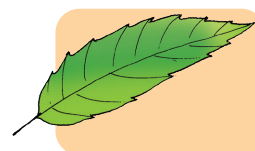
(d)



(e)



(f)



(g)

- Observe the leaves of different plants in your neighbourhood and fill the table.

Draw and name the leaf	Colour	Shape	Texture	Any other observation
 Lemon	Green	Oval	Smooth	Strong smell

Many people have difficulty in making careful observations of the things they see. So, the above type of observation table helps them to focus on specific

observations which may be useful in building upon the concepts further. Two very important aspects of observing are: attention to detail and ability to identify what is relevant to a particular investigation. 'Colour', 'Shape' and 'Texture' are self-explanatory here and focus on specific traits which the teacher wants the children to focus on. The last column which asks the learners to note down any other observations gives scope to individual differences in their observations. Some of these observations may be relevant for the conceptual understanding whereas some may be observations which may not be related to the present context. However, they are important as they may help in future learning and hone observing skills.

Let us understand this further by following possible responses from the children of the class.

Megha - 'The leaf smells like a lemon'

Joshua - The leaves have different types of lines on them.

Syeda - The leaf is very symmetrical.

The teacher can use some of these observations to introduce other concepts like types of venations, patterns in leaves, symmetry etc.

In developing the skill of observation, children learn to use one or more or all senses. When we think of observation, we mostly tend to think about 'seeing' or observing through our eyes. But observations include smelling of things, hearing, tasting, and feeling. Children have different learning styles and make use of different senses and combination of senses in the learning process.

Let's reflect –

- 1) In the above activity, which senses are being used for various observations?
- 2) Is this activity inclusive?

Observations may be qualitative or quantitative. Qualitative observations are descriptive and do not require measurements or reading data (numbers) whereas quantitative observations involve data. The observations in the above activity are qualitative. An example of quantitative observations can be noting the temperature using a thermometer, finding out the hottest data from the data of weekly weather charts etc.

Developing the skill of observation is honed by creating opportunities for children to make a wide range of observations. Discussion plays a significant role in encouraging observations. In the initial stages of development of this skill, talking about their own observations and hearing others help to make a sense of their observations. Gradually they progress to finer observations and focus on conceptual understanding.



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Indicators of Progression in the Skill of Observing-

The learners

- Identify differences and similarities in the materials
- Make use of more than one sense in exploring surroundings or materials
- Extend the range of observations
- Take adequate observations to answer the question in investigation
- Differentiate between observations which are relevant to the task
- Repeat observation to eliminate errors
- Take precautions for accurate observations
- Use instruments like hand lens, thermometer, scale etc to record observations

Teachers can help in the development and progression in skills and ability to gather evidence through observation by

- Providing opportunities to observe like nature walks
- Providing a range of interesting materials and resources
- Providing opportunities to share their observations of the happenings in their daily life
- Allowing sufficient time to observe while doing a task or an activity
- Encouraging children to make a wide range of observations by giving prompts or asking questions

**CHECK YOUR PROGRESS 4.2**

1. A teacher asks the learners to close their eyes and listen to various sounds in the room and write about them. Does this activity promote observing skill?
2. Which of the following tasks promote observing skill in young learners
 - a) Picture comprehension b) Story telling
 - c) Poem recitation d) Colouring
3. Which of the following involve observing?
 - a) Writing similarities and differences between a mango tree and neem tree during a nature walk

- b) Identifying patterns in the tiles of a monument during a field trip
- c) Writing a slogan on water conservation
- d) Writing an essay on water conservation

Answers:

1. Yes 2. a 3. a and b

4.4.2 Classifying

Classifying is one of the important process skills which help in concept acquisition. The ability to distinguish between examples and non-examples is an important aspect of concept development and classification helps in this. The ability to classify or categorise is developmental and does not come spontaneously to children. They must be provided opportunities of sorting on the basis of various criteria and gradually establish relationships between various groups. Usually, classification activities go hand in hand with observation activities. For example, a teacher can extend the above activity on observation of leaves by asking the learners to classify the leaves on the basis of shape, texture or venation. Here, the children initially focus on one attribute only. Gradually they can be facilitated for focussing on more than one attribute. This type of activity helps children in comparing and contrasting between objects and gradually moving towards a complex classification system.

A simple classification may be done on the basis of one attribute at a time.

Example: A teacher asks the students to collect 20 leaves and categorise them on the basis of venation



CHECK YOUR PROGRESS 4.3

1. A teacher lists 20 animals and asks the children to put them into two groups - animals whose ears can be seen and animals whose ears cannot be seen. Which process skills are being promoted in this task?

a. Observing	c. Classifying
d. Both	e. Neither
2. Design a classification activity for nature walks.
3. Rajesh, a Class 4 teacher gives his students a task of conducting a survey to find out the various modes of transport used by the Class 4 children to come to school. Do you think this task can be used to develop classification? Explain how?

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4. How does classifying help in the understanding of a concept? Explain with an example

Answers:

1. c

4.4.3 Predicting

Predicting is another very important process skill which could become a starting point for exploration or investigation by learners. One of the important questions, primary school teachers need to ask in the classroom is ‘What would happen if---?’. One needs to understand that prediction is different from ‘guessing’ and it is a scientific process skill. Guessing usually does not have a logical foundation but prediction is backed reasoning based on previous experiences, observations or a logical thought. Research in children’s thinking indicates that they often do not guess unless pressured for lack of time. Given a chance, they make predictions which in their mind are justified by some kind of experiences even though they may not be able to articulate these connections explicitly.

Refer to activity in section 4.3.1 where the teacher is asking the learners to think if the given object will float or sink. Here, learners may be writing responses based on some of their previous experiences or observations. They may have seen a plastic mug floating in a bucket of water and hence predict that an object made of plastic floats in water.

Let us further understand how prediction helps in concept building through the following activity. This activity is the same as the above but elaborated and extended.

Learning objective- To enable the learners to predict whether given objects float or sink

Rajesh, a Class 4 teacher provides a tray with a number of objects like a metal key chain, a plastic spoon, a coin, an eraser, a sharpener, a pencil, metal spoon, crayon, leaf, a small aluminium paper sheet, a paper boat, a piece of potato. He asked the children to predict which objects will float and which will sink and then to put each of the above objects one by one in a jug of water and note down their observations in the following table.

Sl. No.	Name of the object	Your prediction - float or sink	Your observation- floats or sinks

The activity can be further extended by asking inquiry-oriented questions like “What would happen if I change the shape or size of the object?” Let the children try changing the shapes and sizes like cutting the crayon, changing the shape of a potato piece, crumpling the aluminium sheet etc and write their predictions and observations. The differences between their predictions and observations can be used as opportunities for further discussion, probe into their alternate conceptions and reasoning patterns. For example, if a child predicts that a longer crayon will sink and when it is cut, it will float, the teacher can ask her or him to further explore if the size of the same material makes any difference in floating or sinking. The learner can extend this activity by taking different sizes of crayons. Children may be reluctant to share their predictions for the fear of being wrong. It is here that we as teachers need to encourage them that difference does not matter and this is how we can test our predictions for a scientific inquiry. The teacher can share how scientists also work like this. Examples like the work of meteorologists in weather prediction can help.

Another interesting activity could be asking learners to record minimum, maximum temperature, time of sunrise, sunset for a week from newspapers and then predict the weather patterns for the next few days. This type of prediction activity involves data collection, data interpretation and finding patterns in the data.

Indicators of Progression in The Skill of Predicting

The learners

- Make an attempt to make prediction based on pre conceived ideas
- Recognise that prediction is different from guess
- Use patterns in observations to make predictions
- Explain reasons for predictions
- Justify prediction in terms related concepts

4.4.4 Questioning/Question Raising

Children are naturally curious and raising questions is an important process in their cognitive development. For this reason, it is important that this process skill should be promoted in the primary classrooms. Here it is important to understand that raising questions or question raising as a process skill is not for clarifying doubts or merely being curious but for further exploration or



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investigation. The questions raised should be investigable. So, raising questions in the context of process skills means asking questions that can lead to inquiry (investigable questions) and recognising the kind of questions that can be answered in this way.

Let us understand this through the following activity given on Page 105 in the NCERT Textbook of Class 4.

Sheena asks her students of Class 4 to spin the following objects and note in the table. She asked students not only to write their observations but also to write questions which come to their mind while doing the task or any other related question. An example observation and question are written.

Object	I Observe	I Wonder	I Think
Coin	It spins but as it slows down it begins to shake.	Why does it start to shake as it slows down? What can I do to make it spin for a longer time? Does the size of the coin affect the speed of spinning?	
Pencil			
Bangle			
Wooden cube			
Wooden spin-ner(top)			
Plastic top			

Sheena encourages her students to write the questions or ask these orally. She notes down the questions asked on the blackboard.

Rahul - Why is the wooden top spinning faster than the plastic one?

Sailaja - Why is the pencil spinning only when kept on the table horizontally?

Nazia - What can I do to make my top stop spinning immediately?

Sheena then asks each group to modify the activity and try to find out answers

for each of these questions. One group takes tops made of different materials and sizes and explores how they are spinning and notes down their observations. We can see here that Sheena is encouraging her students to think of questions and further investigate instead of merely asking the students to make observations in the activity.

Though the aim of developing the skill of questioning is to help children raise questions which are investigable, teachers should not discourage other types of questions. Gradually children will begin to recognise and discriminate between questions-which can be investigated and which cannot be investigated at that point. This understanding comes gradually through putting ideas and evidence together. The clarity in children's questions indicate their degree of awareness of what they already know and what they want to know leading to greater conceptual clarity. The role of teacher here is also to help students to convert their questions into investigable or exploratory questions. 'Why a pencil spins and a stone does not spin?' is not an investigable question for primary children whereas "How should a pencil be kept so that it can spin?" is an investigable question.



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Indicators of Progression in Raising Questions

The learners

- Ask a variety of questions
- Participate in discussion about the questions
- Recognise the difference between investigable and non-investigable question
- Suggest a plan of action to find an answer to the question
- Convert their questions into investigable questions
- Recognise that only certain questions can be investigated at that time



CHECK YOUR PROGRESS 4.4

1. Explain how you will know whether your students are merely guessing or using the skill of predicting. Think of some probing questions to be asked to differentiate between the two.
2. Neeta, a Class 4 teacher gives a group task to her students to conduct a water audit in the neighbourhood. Students have to collect information regarding the sources of water, consumption of water in each family, steps taken for water purification and conservation, expenditure

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incurred (water billing), cultural practises related to water. List the process skills being promoted in this activity.

3. Discuss how the skill of question raising is developed through the above activity. Also discuss how you will help the children develop investigable questions.

4.4.5 Communicating

Communication is an important process skill due to its role in developing an understanding of the world around us and making connections. It is often a pathway to express observations, experiences, thinking and learning and in doing so in a cohesive manner, children's other process skills are also developed. It is important to understand that communication is not merely reporting observations or any other experiences. Rather both written and oral recording and communicating are an integral part of the activity.

Extending the range or modes of communication requires knowledge of the ways of presenting information. For example, using graphs to communicate observations requires different skills and conventions. Increased use of graphs, diagrams, flow charts, symbols, drawings is an important developmental process in learning. Communicating also involves the ability to interpret data. The ability to interpret data, search for patterns, represent information in multiple forms like graphs, tables, flow charts etc are hallmarks of effective recording and communicating. Communicating also encompasses various oral forms like theatre, role play, discussion etc.

The gradual use of scientific vocabulary is also part of progression in the skill of communication. For example, children in earlier classes may say that water 'disappears' from a wet cloth when put in the sun and replaced with 'evaporates' at later stages.

It is also important to understand that communication is a two-way process and involves listening to others, looking for information from other sources which help in fostering clear expression of ideas and healthy respect for other's ideas.

Indicators for Progression in Communicating

The learners

- Talk freely about their activities and ideas
- Listen to other's ideas
- Write short descriptive reports
- Use drawings, tables, graphs to record and communicate their observations
- Use appropriate scientific language

**CHECK YOUR PROGRESS 4.5**

1. Which graphical representations are suited for learners at the preparatory and foundational stage to communicate their observations and findings?
2. State True or False.
 - a. Interpreting weather data from newspapers is a communication skill.
 - b. Communications also means listening to others.

Answers:

2. a) False b) True

4.5 PLANNING INVESTIGATIONS: INTEGRATED PROCESS SKILLS

This is an integrated process skill as it encompasses many process skills like observing, classifying, communicating, measuring, inferencing, controlling variables. You will recall that we have discussed in the previous section, how a teacher can help the children in framing investigable questions. This is a step further. Young children may not be able to think through a series of actions or plan and need scaffolding by the teacher after each step i.e. they need to see what happens as a result of the first step before thinking about what to do next. Children should be encouraged to think along the lines of 'fair testing' which helps to develop the process skill of identifying and controlling variables. Let's understand this through the following example:

Children in Bhavneet's class want to explore if seeds germinate in the same manner when watered with different amounts of water i.e. to investigate

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the effect of water on germination of seeds. Here the teacher can help in developing the skill of ‘identifying and controlling variables’ by drawing their attention for fair testing i.e. need for establishing the same basis for comparison. In this case seeds, temperature, and duration of sunlight are same (constant) but the amount of water (variable) is varied. Some probing questions can be asked:

Can we take different types of seeds and water them with different amounts of water? (for introducing to fair testing)

What are the things we need to keep the same and what are the things which we can change? (for introduction to “identifying and control of variables”)

Let’s reflect - In this activity, discuss how the process skill of observing, predicting, questioning, communicating and measuring are used and promoted.

Indicators for Progression in Skills in Planning and Conducting Investigations

The learners

- Frame an investigable question and suggest action plan
- Identify variables that have to be changed and kept same for fair testing
- Devise steps for the same
- Gather resources and materials
- Implement plan of action
- Identify patterns in their observations and data
- Draw conclusions or inferences and try to explain the reasons in appropriate vocabulary
- Recognise that there may be more than one explanation for the evidence and conclusions can be tentative

What can the teachers do to encourage investigations and experimentation in the classroom?

- Providing opportunities to children for exploring their questions
- Discussing their plan of action and providing support at various stages
- Scaffolding for fair testing and control of variables in investigations / experiments
- Reviewing and providing feedback at appropriate stages

**CHECK YOUR PROGRESS 4.6**

1. List the basic process skills involved in designing an investigation or experimentation.
2. In an activity designed to investigate the factors affecting the solubility of various substances in water, what variables should be kept the same and what should be changed?
3. In an investigation/activity designed to find out how bread gets spoilt, what are the various process skills involved?
4. Which of the following tasks are best suited to introduce the process skill of identifying and controlling variables?
 - a. Finding about factors affecting shadow formation
 - b. Finding information about various professions in the neighbourhood
 - c. Visiting an animal shelter
 - d. Interviewing grandparents about festivals

Answers:

4. a

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4.6 DESIGNING PROCESS SKILL-BASED ACTIVITIES

In the previous sections, we have discussed some process skills which are important in primary classrooms, especially in understanding the concepts in the curriculum of 'The World Around Us'. We have also analysed a few activities given in the NCERT Textbook 'Our Wondrous World' in terms of process skills. Let us now understand how to design some activities suitable for development of one or more process skills.

An activity can involve one or more process skills. While designing activities, some of the questions /pointers we need to keep in mind are-

- What are the learning outcomes?
- What are the process skills that can be developed and how?
- What resources/materials are needed?
- What are the steps to be followed?
- Is it group or individual activity?

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- How will the students communicate?
- What is the purpose of the activity?

Suggestive Activity Template

1. Grade
2. Topic
3. Number of students
4. Learning outcomes
5. Process skills
6. Materials needed
7. Instructions
8. Observation table

**CHECK YOUR PROGRESS 4.7**

1. Design an activity to promote process skills in Class 4 in the unit 'Health and Wellbeing'. Write the learning outcomes and mention the process skills.
2. A teacher asks learners to collect various currency notes and write the similarities and differences. Which process skills can be developed through this task?
 - a. Observing, communicating, inferencing
 - b. Observing, classifying, inferencing
 - c. Communicating, predicting, measuring
 - d. Questioning, observing, measuring

Answers:

2. a

4.7 MAPPING

Mapping in the primary classes provides a basis for understanding the world around us and develops spatial awareness. It involves spatial thinking, visualisation and reasoning. Mapping skills include understanding of relative positions, directions and scale. It involves children reading and interpreting maps, understanding of directions, symbols and proportional scaling. Mapping involves both reading maps and making maps. Mapping as a child's process skill in the primary classes follows a 'local' to 'global' approach, starting from mapping of the surroundings like home, classroom and gradually moving to

larger spaces like the school, neighbourhood, city, country and the world.

The following Fig. 4.2 illustrates the key concepts in mapping at primary level.



Fig. 4.2: Key Concepts of Mapping

As children in the primary classes are at the pre operational and concrete operational stages of development and not yet ready with proportional reasoning, they cannot draw maps as per the scale. It is therefore best to develop mapping skills in the sequence given below

1. Drawing one's surroundings
2. Using symbols for self-drawn maps
3. Interpreting maps (neighbourhood, school etc)
4. Interpreting standard maps (cities, country etc)
5. Drawing maps without proportionality (unscaled maps)
6. Drawing and interpreting scaled or standard maps

Let us analyse an activity given in the NCERT textbook 'Our Wondrous World' on page 30, Chapter 2 in terms of developing mapping skills.

Activity 4

Draw a map showing the route from your school to the public library or any such place near your school. Identify landmarks like parks, shops, hospitals, bus stops, etc. on the map.



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Let's reflect –

- i) How is this activity promoting spatial thinking?
- ii) What other process skills are involved?
- iii) Explain how symbols and legends can be introduced in this activity.
- iv) How will you introduce mapping to a visually challenged student?

Mapping can be integrated with the teaching -learning of other topics also. For example, if you are teaching about plants, you can ask the learners to map a garden or park to represent a variety of plants. You can show on maps, where particular plants or crops are grown. Similarly, while teaching about transportation, you can use resources like train route maps, local city maps etc.

Developing mapping skills from the early years not only promotes visualisation and spatial thinking but also critical thinking and problem solving in the later years.

**CHECK YOUR PROGRESS 4.8**

1. Jyoti, a Class 3 teacher, asks her students to draw a map of their school. What are the process skills involved in this activity?
2. How do you think mapping activities differ across Classes? How is a map drawn by a Class 2 student different from a map drawn by a Class 5 student?
3. What are some of the challenges experienced by the children of primary Classes in mapping?
4. Do you think standard maps like a map of India or the World can be introduced in Class 3?

4.8 STRATEGIES FOR PROCESS SKILL DEVELOPMENT

In the previous sections we have discussed various process skills which are important for teaching learning at the foundational and preparatory stages in school. We have also discussed the role of the teacher in development of these process skills and how activities can be specifically designed to promote them. Following are some common strategies you can use for helping process skill development in young learners

- Providing opportunities for children to explore materials and phenomenon directly

- Creating opportunities to share guided and unguided observations
- Facilitating group discussions for sharing ideas
- Providing opportunities for various investigations
- Scaffolding for raising investigable questions
- Encouraging children to check that their ideas are consistent with evidence and observations
- Helping children in recording systematically
- Teaching techniques needed for advance skills in using instruments
- Providing access to alternative procedures
- Encouraging critical reflection on how they have learnt
- Using cueing and questioning to draw the children's attention to a particular indicator within a skill or learning material



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A teacher can use questioning to encourage the use of process skills

What would you like to know about _____

Why do you think will happen if _____

What difference do you notice _____

What are the similarities in _____

What do you think A is growing better than B?

How can we find out if _____

Did you notice any connection between _____

How will you share your ideas?

What will you do to find out _____

4.9 LET US SUM UP

- Process skills are a range of skills needed for exploring and developing conceptual understanding.
- Process skills form the core of inquiry based, hands-on learning in primary classrooms.
- Learners use process skills by asking questions, making observations, collecting data, taking measurements, inferencing data, predicting the

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outcomes of manipulating one variable, developing investigations and communicating their findings to others.

- Important process skills which need to be developed in young learners are observing, classifying, predicting, questioning, communicating, mapping and experimenting.
- Teachers can promote development of process skills in learners by designing activities and tasks.

4.10 END QUESTIONS

1. Process skills are important for teaching-learning of TWAU (The World Around Us). Which of the following is not a process skill?
 - a. Observing
 - b. Predicting
 - c. Memorising
 - d. Inferencing
2. A teacher gives a list of 10 food items and asks them to put these into two groups-1) which get spoiled very quickly (a day or two) and 2) which can stay fresh for a long time (a week or more). Which process skills are being promoted in this task?
 - a) Observing, measuring, questioning
 - b) Observing, classifying, predicting
 - c) Observing, inferencing, classifying, predicting
 - d) Observing, measuring, inferencing
3. A teacher asks her students to grow a flowering plant and note down the time taken for bud formation, for the bud to grow into a flower and then flower to dry up. Which process skills are involved in this activity?
4. In order to promote mapping skills in primary school children, what should be the sequence of activities—showing standard maps, interpreting neighbourhood maps, drawing surroundings, drawing according to scale.
5. Design an activity to promote the following skills
 - a) Observing
 - b) Predicting
 - c) Experimenting

4.11 REFERENCES

1. NCERT. (2024). Our Wondrous World. A Textbook for Class III. The World Around Us.
2. NCERT. (2025). Our Wondrous World. A Textbook for Class IV, The World Around Us.
3. NCERT. (2025). Our Wondrous World. A Textbook for Class V, The World Around Us.

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Notes